Some basic game theory

ECN 714

Bayesian games

Introduction

- In many games of economic interest, players may have imprecise information about some aspects of the game.
- For instance, a player may not be certain about her oponents' exact payoffs.
- A ``representation'' of the strategic situation described is necessary.

Example

- Players 1 and 2 choose actions simultaneously and independently; Player 1 chooses Up or Down, player 2 chooses Left or Right.
- Player 2 knows the game's payoffs.

	L	**R**
U	1, 2	0,1
D	0, 4	1,3

- Player 1 knows her own payoffs but she is not certain about 2's payoffs.

Player 1 as a Bayesian player

- Player 1 knows his own payoffs but she is not certain about 2's payoffs.

	L	**R**
U	1	0
D	0	1

- Player 1 believes that player 2's payoffs are either

	L	**R**			**L**	**R**
U	2	1	or	U	3	4
D	4	3		D	1	2

and assigns probability 0.6 to the first alternative and 0.4 to the second one.

- Combine payoffs in a table, let states be $t = a$ and $t = b$ and player 1's *beliefs* are $p_a = 0.6$ and $p_b = 0.4$.

$t = a, p_a = .6$

	L	R
U	1, 2	0,1
D	0, 4	1, 3

$t = b, p_b = .4$

	L	R
U	1, 3	0,4
D	0, 1	1, 2

Benchmark: complete information

- Complete information: both players know whether $t = a$ or $t = b$.
- Strategies are functions:

$[s_1(a) = U, s_1(b) = D]$, $[s_2(a) = L, s_2(b) = R]$

- Solution:

If $t = a$ then a NE is (U, L).

If $t = b$ then a NE is (D, R).

- If listed probabilities are correct, an outside observer will see 60% of the time (U, L) and 40% of the time (D, R).

Incomplete information

- Player 1 does not observe t; player 2 does. From 1's viewpoint, 2's strategies are functions.
  - If $t = a$, 2 plays L because L is dominant.
  - If $t = b$, 2 plays R because R is dominant.
- Player 1 has two strategies U and D. Single information set.

$$u^1(U) = 0.6 \times 1 + 0.4 \times 0 = 0.6$$
$$u^1(D) = 0.6 \times 0 + 0.4 \times 1 = 0.4$$

Hence, player 1 chooses U.

Conclusions

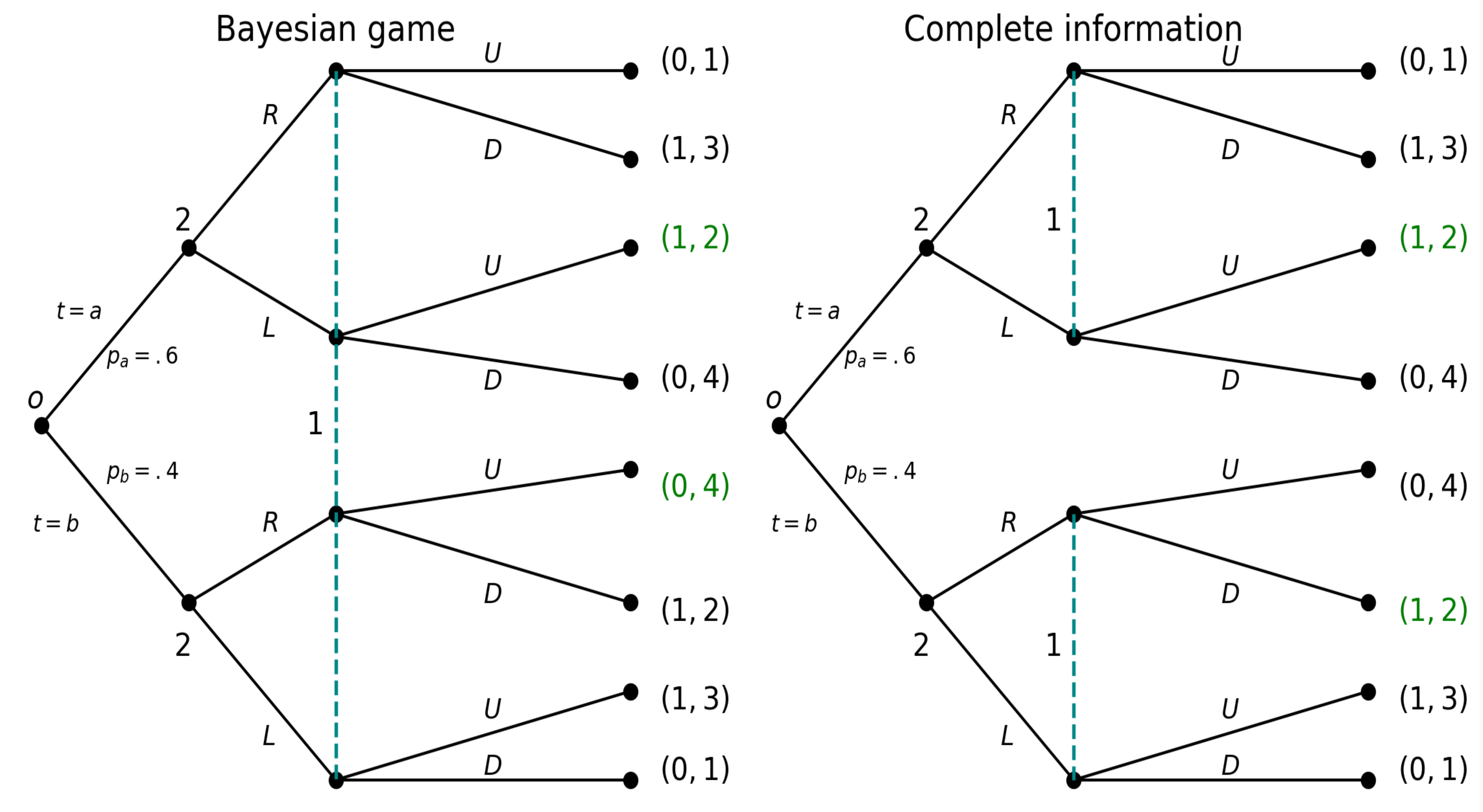
- The intuitive outcome of the game was identified.
- Under incomplete information, since player 1 does not know the state t, she can no longer change her action depending on it.
- An outside observer would see player 1 choosing U every time, and player 2 changing so that 60% of time he plays L and 40% of the time he plays R.
- The incomplete information game is *not* a combination of the two complete information games.

Bayesian games

- Goal: to obtain a game representation and a solution concept to generalize the example.
- Incomplete information is modeled by adding a move by nature before the game begins to the extensive form representation.
  Nature thus determines the realization of players' private information.
  Harsanyi (1967), Harsanyi (1968a), Harsanyi (1968b).
- Uncertainty about payoffs or other elements *is a move by nature that takes place at the root of the game.*

- To model a player that is not certain about its opponents, a Bayesian game creates types for the opponents (not for the player that lacks the information.) Each type represents a possible realization of the information.
- A Bayesian game is the normal form representation of the game *played* before the private information is revealed.
- A Bayesian Nash equilibrium (BNE) is a Nash equilibrium of the normal form game described.

Harsanyi's idea applied to the example



Bayesian games–definitions

Definition 1 (Private information) Private information is the information that an agent has that is not available to any other agent.

- Intuitively in a Bayesian game,
  players first observe their private information or type.
- Players have beliefs about opponents' types.
  In the example, player 1's beliefs are that player 2 has payoffs "a" with probability 0.6 and payoffs "b" with probability 0.4.
- Each player simultaneously and idependently chooses his or her action.
- Final payoffs are determined by realization of information and actions chosen.

Definition 2 (Bayesian game) A Bayesian game Γ $= [\mathcal{J}, \{T_i, A_i, U_i, \beta_i\}_{i \in \mathcal{J}}]$ is a collection of

1. $\mathcal{J}$, finite set of players.
2. T_i, set of player i's private information or types.
3. A_i, set of actions available to player i.
4. $U_i : T \times A \longrightarrow \mathbb{R}$, player i's payoff function, where $T = \prod_{i \in \mathcal{J}} T_i$ and $A = \prod_{i \in \mathcal{J}} A_i$.
5. $\beta_i : T_i \to M(T_{-i})$, player i's beliefs about the other players' private information.

$M(T_{-i})$ is set of probability distritubions over T_{-i}.

- When T_{-i} is finite, $\beta_i(t_{-i}|t_i)$ is the probability that i assigns to t_{-i} when i's information is t_i.
- More generally, $\beta_i(\cdot|t_i)$ is a probability distribution on T_{-i} for each t_i.

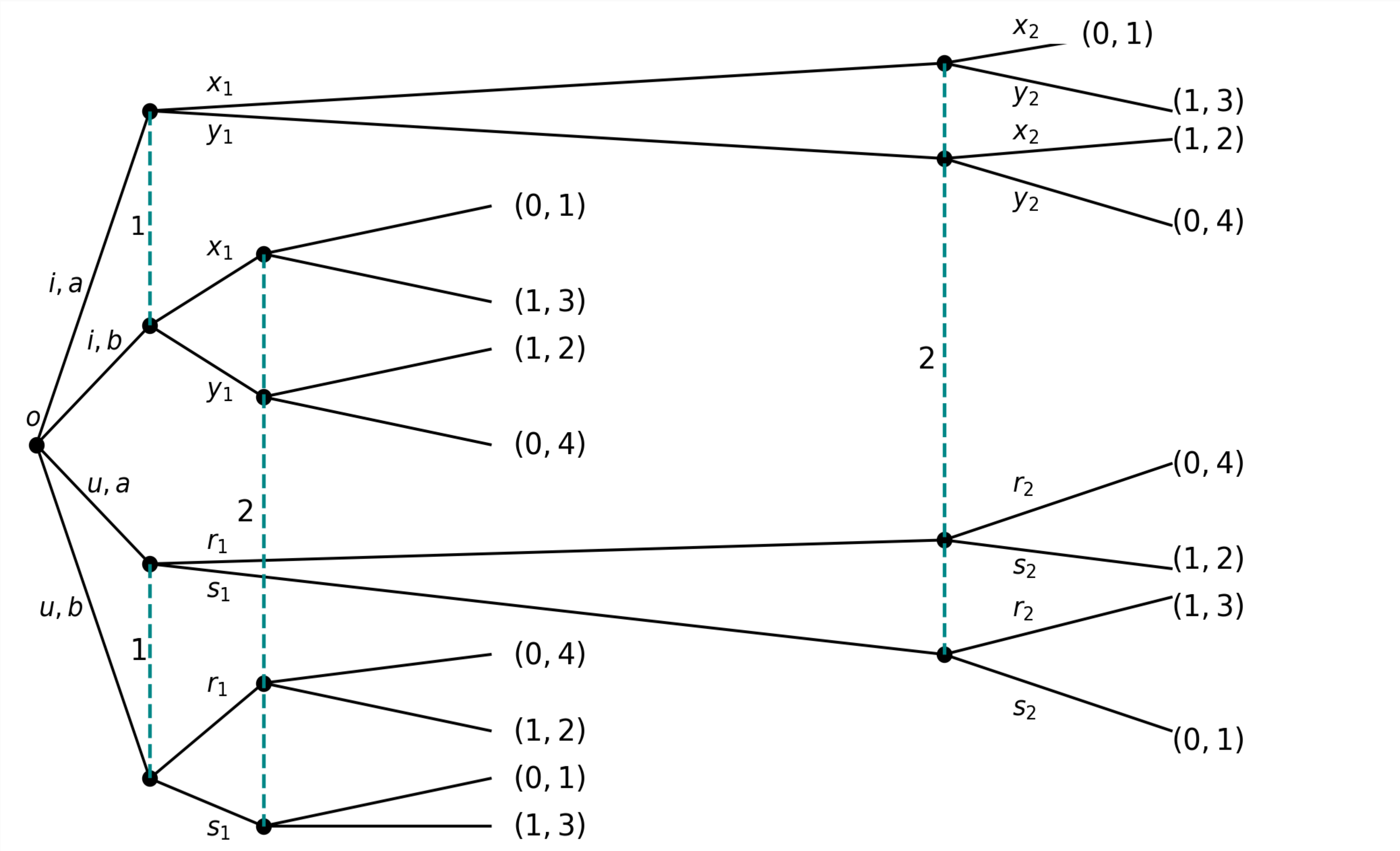
Beliefs

Definition 3 (Consistent beliefs) Beliefs are consistent with a probability P on T if for all $i \in N$, and for all $(t_{-i}, t_i) \in T$,

$$\beta_i(t_{-i}|t_i) = \frac{P(t_{-i}, t_i)}{\sum_{t_{-i} \in T_{-i}} P(t_{-i}, t_i)}.$$

Definition 4 (Common prior) When beliefs are consistent with a given probability distribution P, it is said that players have a common prior P.

Beliefs and common prior



Two players, $T_1 = \{i, u\}$, say informed or uninformed; and $T_2 = \{a, b\}$.

Then,

$P(t_1, t_2)$	$t_2 = a$	$t_2 = b$	$\sum_{t_2 \in T_2}$
$t_1 = i$	$P(i, a)$	$P(i, b)$	$P_{T_1}(i)$
$t_1 = u$	$P(u, a)$	$P(u, b)$	$P_{T_1}(u)$
$\sum_{t_1 \in T_1}$	$P_{T_2}(a)$	$P_{T_2}(b)$	1

$$\sum_{t_1 \in T_1,\ t_2 \in T_2} P(t_1, t_2) = 1$$

Example: Consistent beliefs

	a	b	$\sum_{a,b}$
i	0	0	0
u	0.6	0.4	1
$\sum_{i,u}$	0.6	0.4	

$$\beta_2(t_1 = i|a) = \frac{P(i,a)}{P(i,a) + P(u,a)} = \frac{0}{0 + .6} = 0$$

$$\beta_2(t_1 = i|b) = \frac{P(i,b)}{P(i,b) + P(u,b)} = \frac{0}{0 + .4} = 0$$

$$\beta_1(t_2 = a|i) = \text{not defined by Bayes rule.}$$

$$\beta_1(t_2 = a|u) = \frac{P(u,a)}{P(u,a) + P(u,b)} = \frac{.6}{1} = .6$$

$$\beta_1(t_2 = b|u) = \frac{P(u,b)}{P(u,a) + P(u,b)} = \frac{.4}{1} = .4$$

Beliefs that are not consistent

	a	b	$\sum_{a,b}$
i	0.06	0.04	0.10
u	0.54	0.36	.90
$\sum_{i,u}$	0.6	0.40	

$$\beta_1(t_2 = a|i) = 0.6$$
$$\beta_1(t_2 = b|i) = 0.4$$
$$\beta_1(t_2 = a|u) = 0.54/0.9 = 0.6$$
$$\beta_1(t_2 = b|u) = 0.36/0.9 = 0.4$$

Strategies

Fix a Bayesian game Γ.

- A *pure strategy* s_i for player i is any function $s_i : T_i \to A_i$. The set of player i's pure strategies is denoted by S_i.
- A *mixed strategy* for player i is a probability distribution $\mu_i \in M(S_i)$, the set of probability distributions on S_i.
- A *behavior strategy* for player i is any function $\sigma_i : T_i \to M(A_i)$. The set of player i's behavior strategies is denoted by Σ_i.
- A *strategy profile* a strategy profile is a collection of strategies, one for each player in the game.

Observations

Let Γ be a Bayesian game with two players.

A pure-strategy profile (s_1', s_2') is a Bayesian Nash equilibrum (BNE) if $(\forall i \in \mathcal{I})$ $(\forall t_i \in T_i)$ $(\forall s_i \in S_i)$ $(\forall j \neq i)$,

$$\sum_{t_j \in T_j} u_i(t_j, t_i, s_j'(t_j), s_i'(t_i))\beta_i(t_j|t_i) \geq \sum_{t_j \in T_j} u_i(t_j, t_i, s_j'(t_j), s_i(t_i))\beta_i(t_j|t_i). \tag{1}$$

- Note that $s_i(t_i)$ could be any element of A_i

Observations 2

Thus equivalently,

$$\sum_{t_j \in T_j} u_i(t_j, t_i, s'_j(t_j), s'_i(t_i))\beta_i(t_j|t_i) \geq \sum_{t_j \in T_j} u_i(t_j, t_i, s'_j(t_j), a_i)\beta_i(t_j|t_i), \forall a_i \in A_i.$$

- Notation: Using expectations

$$E[u_i|t_i, s'_i, s'_j, \beta_i] = \sum_{t_j \in T_j} u_i(t_j, t_i, s'_j(t_j), s'_i(t_i))\beta_i(t_j|t_i).$$

Observations 3

Let

$$E[u_i|s_i', s_j'] = \sum_{t_i \in T_i} \sum_{t_j \in T_j} u_i(t_j, t_i, s_j'(t_j), s_i'(t_i)) \beta_i(t_j|t_i) P_{T_i}(t_i)$$

where P_{T_i} is the marginal of P on T_i.

- Then, (s_i', s_j') is a BNE if

$$E[u_i|s_i', s_j'] \geq E[u_i|s_i, s_j'], \forall s_i \in S_i. \tag{2}$$

- The inequality in Equation 1 implies the inequality in Equation 2. When $P_{T_i}(t_i) > 0$ for every t_i, both inequalities are equivalent.

Dominant strategy

- A pure strategy s_i is (weakly) dominant for player i in Γ, if

$$\forall s_{-i}, E[U_i|s_{-i}, s_i] \geq E[U_i|s_{-i}, s_i'], \forall s_i'$$

- A profile $s = \{s_i\}_{i=1}^{I}$ is a (weakly) dominant strategy equilibrium if s_i is (weakly) dominant for $i = 1, \dots, n$.

Baysian Nash equilibrium

Definition 5 (Bayesian Nash equilibrium) A behavior strategy profile $\sigma' = (\sigma'_1, \ldots, \sigma'_N)$ is a Bayesian Nash equilibrum (BNE) if $(\forall i \in \mathcal{I})$ $(\forall t_i \in T_i)$ $(\forall \sigma_i \in \Sigma_i)$,

$$E[u_i | t_i, \sigma'_i, \sigma'_{-i}, \beta_i] \geq E[u_i | t_i, \sigma_i, \sigma'_{-i}, \beta_i].$$

Note that

$$E[U_i|t_i,\sigma_i,\sigma'_{-i},\beta_i] = \sum_{t_{-i}\in T_{-i}}\sum_{a_1\in A_1}\cdots\sum_{a_I\in A_I} u_i(t_{-i},t_i,a_1,\ldots,a_I)$$
$$\Big(\prod_{j\neq i}\sigma'_j(a_j|t_j)\Big)\,\sigma_i(a_i|t_i)\beta_i(t_{-i}|t_i).$$

Universal type space

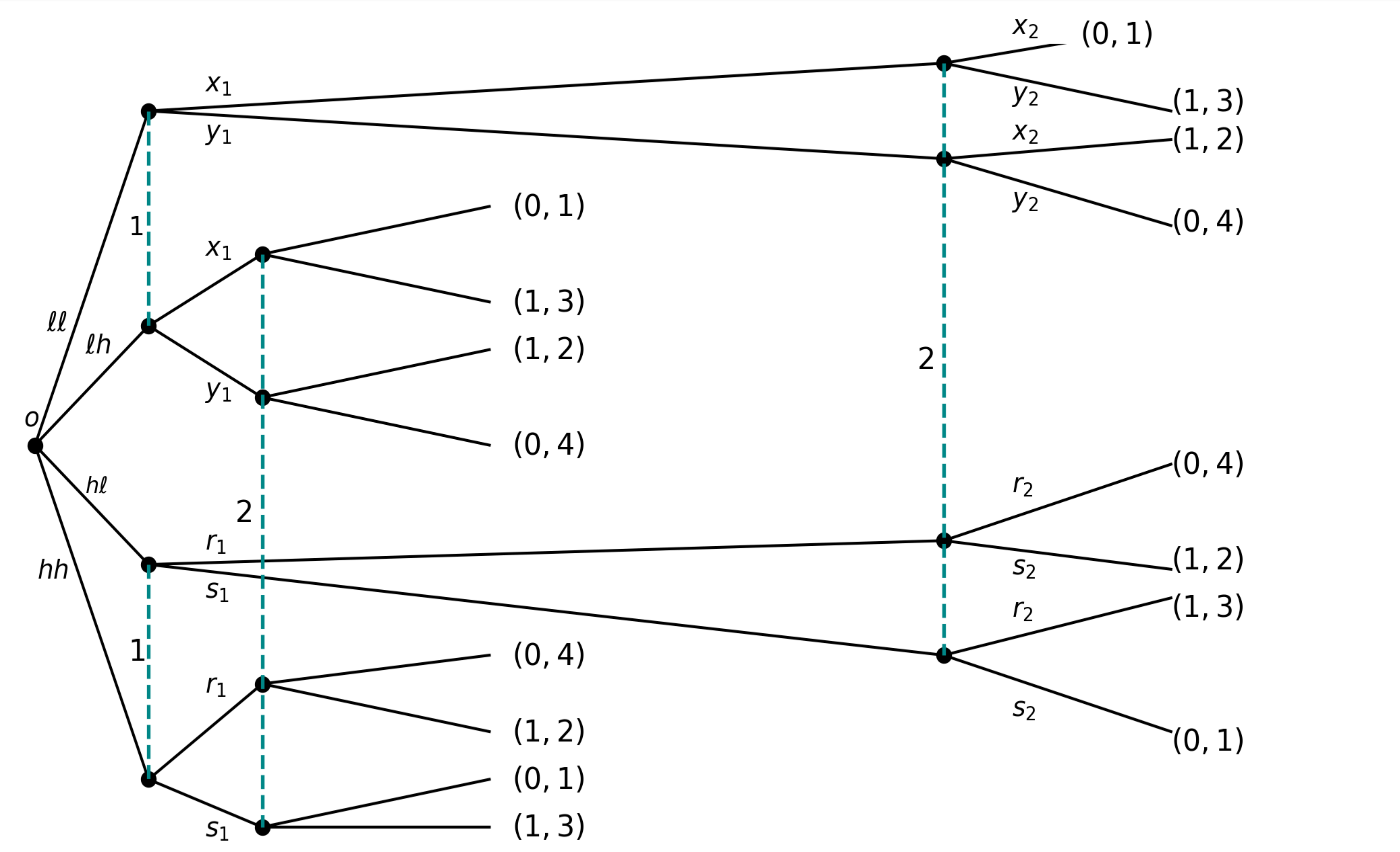
Difficulty with incomplete information. Refer to leading example.

- Player 2 observes t_2, player 1 does not. Player 1 must have beliefs about t_2.
- Depending on 1's beliefs, 2's preferred strategy may change.
- In the example, 2 has a dominant strategy in each game where a game is defined for each realization of t_2. So this is not an issue. It is in general.

- Player 2 does not observe 1's beliefs; 2 must form beliefs about player 1's beliefs.
- Player 1 optimal action may depend on player 2's choices. In turn these choices may depend on player 2's beliefs. Thus player 1 must form beliefs, about player 2's beliefs about player 1's beliefs.
- Continue ad infinitum.

See Brandenburger and Dekel (1993)} and Mertens and Zamir (1985).

Example: auction, stylized representation



Independent private values model (IPVM)

A standard model in trading environments.

- A seller has a single, indivisible object.
- There are I potential buyers or bidders.
- Bidder i privately observes her valuation $x_i \in [0, 1]$ for the object.
- Preferences: If i receives the object and pays an amount m, i's payoff is $x_i - m$. Expected utility is assumed.
- Beliefs: Variables x_i are i.i.d.. Each x_i has density f, assumed to be strictly positive and with full support. This is also a symmetry assumption, i.e. bidders are ex ante equal.

Notation

- $x = (x_1, \ldots, x_I)$, $x_{-i} = (x_1, \ldots, x_{i-1}, x_{i+1}, x_I)$.
- $X_1, X_2, \ldots, X_I$ sets, $X = \prod_{i=1}^{I} X$.
- $M(X_i)$ is the set of probability distributions on X_i.
- First order statistic: $x^{(1)} = \max \{x_i\}_{i=1}^{I}$;
- Second order statistic: $x^{(2)}$

Classic auctions

- Sealed-bid, second-price auction (SPA).
  - Bidders simultaneously and independently submit their bids in sealed envelopes.
  - Envelopes are collected and opened.
  - Highest bidder, the winner, is assigned the object. Ties in determining the highest bidder are resolved according to some predetermined procedure.
  - The winner pays the second-highest bid.
  - Strategies?

Second-price, sealed-bid auction (SPA)

Theorem 1 (SPA) IPVM. The strategy profile $\{b_i(x_i) = x_i\}_{i=1}^{I}$ is a weakly dominant-strategy equilibrium (WDSE) of the SPA.

Proof. Let b_{-i} be the vector of i's oponents bids.

$$u_i\left(x_i, b_i, b_{-i}^{(1)}\right) = \begin{cases} 0 & \text{if } b_i < b_{-i}^{(1)} \\ \left(x_i - b_{-i}^{(1)}\right) & \text{if } b_i > b_{-i}^{(1)} \\ \left(x_i - b_{-i}^{(1)}\right) P(tie) & \text{if } b_i = b_{-i}^{(1)} \end{cases}$$

- $x_i < b_{-i}^{(1)} \implies i's \text{ best response is to lose}$, thus $b_i = x_i < b_{-i}^{(1)}$.
- $x_i > b_{-i}^{(1)} \implies i's \text{ best response is to win}$, thus $b_i = x_i > b_{-i}^{(1)}$.
- $x_i = b_{-i}^{(1)} \implies i \text{ is indifferent}$, thus $b_i = x_i$. QED

SPA-Remarks

- Weakly dominant strategy equilibrium (WDSE) means a strong recommendation.
  Principle: winner's payment does not depend on winner's bid.
- Result is stronger than WDSE of the implicit game. (Continuum technicality.)
- Where is independence used? And symmetry?
- Are there other BNE?
- Solution is efficient.

Classic auctions

- Sealed-bid second-price auction (SPA)
- Sealed-bid first-price auction (FPA): Bidders submit their bids in sealed envelopes simultaneously and independently. Envelopes are opened. Highest bidder gets the object and pays his or her own bid.
- English auction
- Dutch auction

Existence of BNE

Finite Bayesian games. Existence of BNE is a consequence of the general existence of NE in finite games.

Existence proofs use a fixed point argument.

- Primitive payoff function: $U_i(x_1, x_2, a_1, a_2)$
- Behavior strategy: $\sigma_i : X_i \to M(A_i)$
- Best response correspondence: $\psi_i(\sigma_{-i}) = \arg\max_{\sigma_i} E[U_i|\sigma_i, \sigma_{-i}]$
- $\psi(\sigma) = \prod_{i=1}^{I} \psi_i(\sigma_{-i})$.

- Theorem of the Maximum (Berge): Continuity of primitive function implies upper-hemicontinuity of the argmax.
  In this case, continuity of $E[U_i|\sigma_1, \sigma_2]$ implies uhc of ψ.
  See for instance, Aliprantis and Border (2006), page 569.
- uhc and convex-valued ψ plus compactness of domain $\implies$ fixed point, therefore existence.

BNE in infinite games

- No general existence result for BNE in infinite games, games where type or action spaces have the cardinality of the continuum.
- Advantages of continuum model
  - Modeling. Perhaps more appropriate, bids and uncertainty. Bids in practice are discrete. Should results depend on whether permitted bids are in pennies or nickels?
  - Technical.
    - Avoid complications of multiplicity of equilibria from mixing between discrete alternatives.
    - Calculus.

BNE in infinite games

- Difficulty with continuum model.
- $E[U_i|\sigma_1, \sigma_2)] = \int_{X_1 \times X_2} \int_{A_1} \int_{A_2} U_i(x, a) \, \mathrm{d}\sigma_1(x_1) \, \mathrm{d}\sigma_2(x_2) \, \mathrm{d}\mu$
- Topology (metric) that makes ψ uhc, has too many open sets to make the domain of ψ compact and viceversa.
- Additional questions.
- Existence of pure strategy equilibria or purification of mixed strategy equilibria.
- Approximation discrete-continuum.

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